

AI Prompting Cheat Sheet

How to get better answers for scientific programming and coding

1. Ask Good Questions

The quality of the answer depends strongly on the quality of the question. Clear and structured prompts produce clearer and more useful answers.

- **Be specific**

Avoid vague questions. Clearly define what you want.

Bad: *Explain forcefields*

Better: *What is a force field in the context of molecular dynamics simulations?*

- **State your knowledge level**

Tell the AI how much background knowledge you have so it can adjust the explanation.

Examples:

– *I am a beginner in Python. Explain the solution step by step.*

– *I am a Master's student in physics. Give me an expert level answer.*

- **State your purpose and define the AI's role**

Explain your situation and what you want the AI to help you with.

Example:

I am a physics student attending a computational physics lecture. I have written simulation code in python, but I am getting errors that I do not understand. Your task is to help me debug my code.

- **Clearly define inputs and outputs**

When asking for code, specify what goes in and what should come out.

Example:

*Write a function that takes **particles** as input, an array of shape $N \times 3$, where N is the number of particles and each row contains the particle position in 3D space. The output should be the updated particle positions after one Euler integration step. Use a timestep of $dt=0.1$.*

- **Allow the AI to ask for missing information**

Prevent wrong assumptions by explicitly allowing clarification.

Example:

If important information is missing, ask me before answering. Do not guess or invent missing details. State your assumptions clearly.

- **Define what must remain unchanged**

When modifying code, specify what should stay the same.

Example:

Do not change the function names, input format, or output format. Only fix the error in the implementation.

2. What would you do?

Imagine you have to design a small pdf document for students who learn about the mean field theory of the Ising model in their final year of their Bachelor's degree. The document should be a small reference for students to self study and understand the derivations up to the calculation of critical exponents. What do you include? How would you instruct AI to create a latex document for you?

I need to create a pdf file for students finishing their Bachelor degree discussing the 3d Mean field of the Ising model. Your task is to do the derivation and create the latex document. 1. Derive the critical exponents of the Ising model using mean field theory. Start with the Hamiltonian for the nearest-neighbor Ising model and decouple the Hamiltonian using mean field to arrive at the effective mean field Hamiltonian. Based on this, calculate the partition sum to derive the self-consistency equation and solve it for the critical temperature. Discuss the behaviour at zero magnetic field and derive the exponent for magnetization based on the selfconsistency equation. Do the same for the susceptibility. Explain each step in the derivation carefully, do not skip over steps or guess if you do not know how to continue. 2. Afterwards, based on your derivation, create a summary of the derivation in a latex style document, which can be used by students to self study this derivation. The document should contain the starting equation and should guide the reader through each step with additional explanation to enhance physical intuition and should lead to the critical exponents in the end. Explain each step carefully. Keep it brief, max 5 pages. Give a complete latex code and include all packages, so I can copy and paste into overleaf.